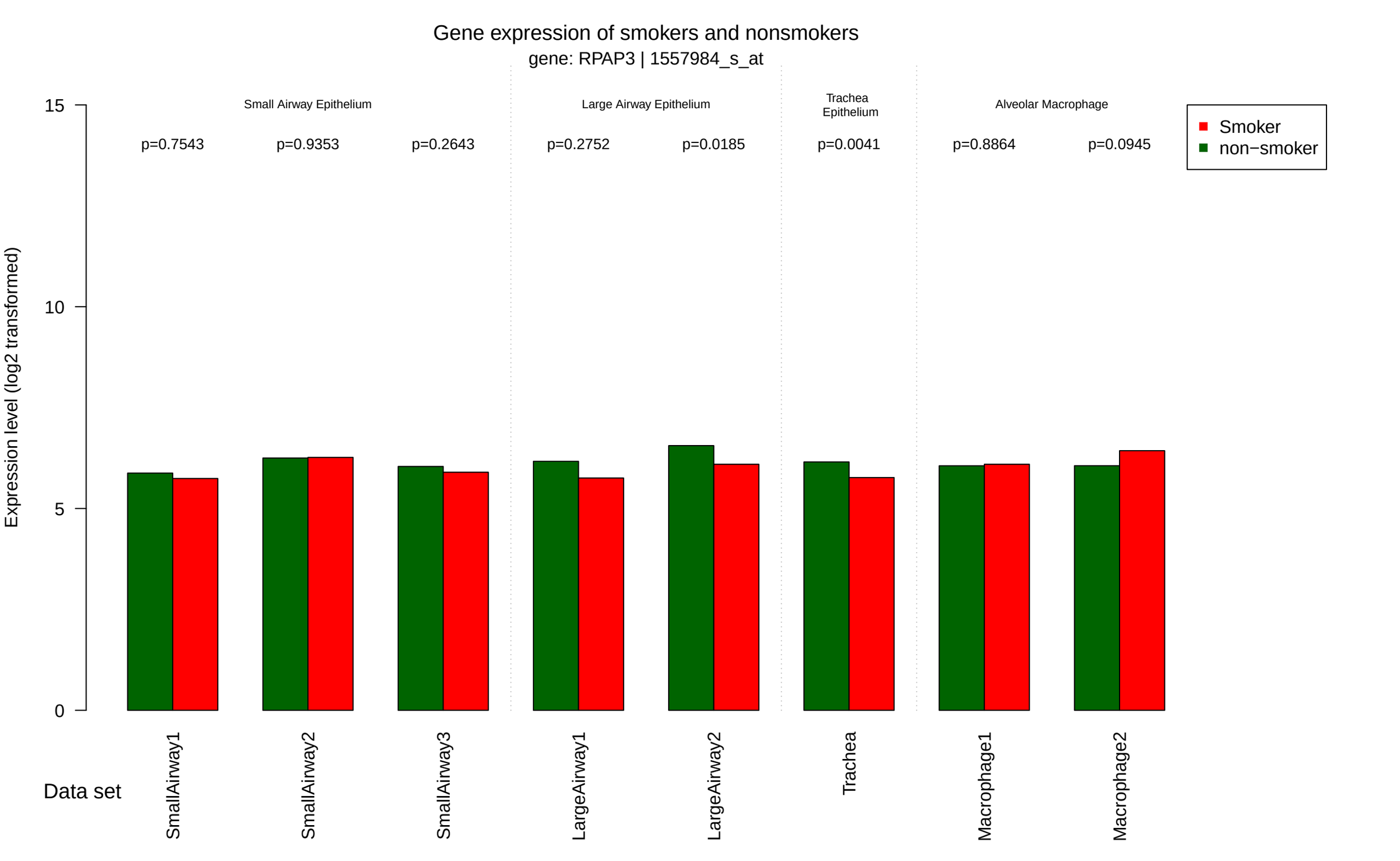
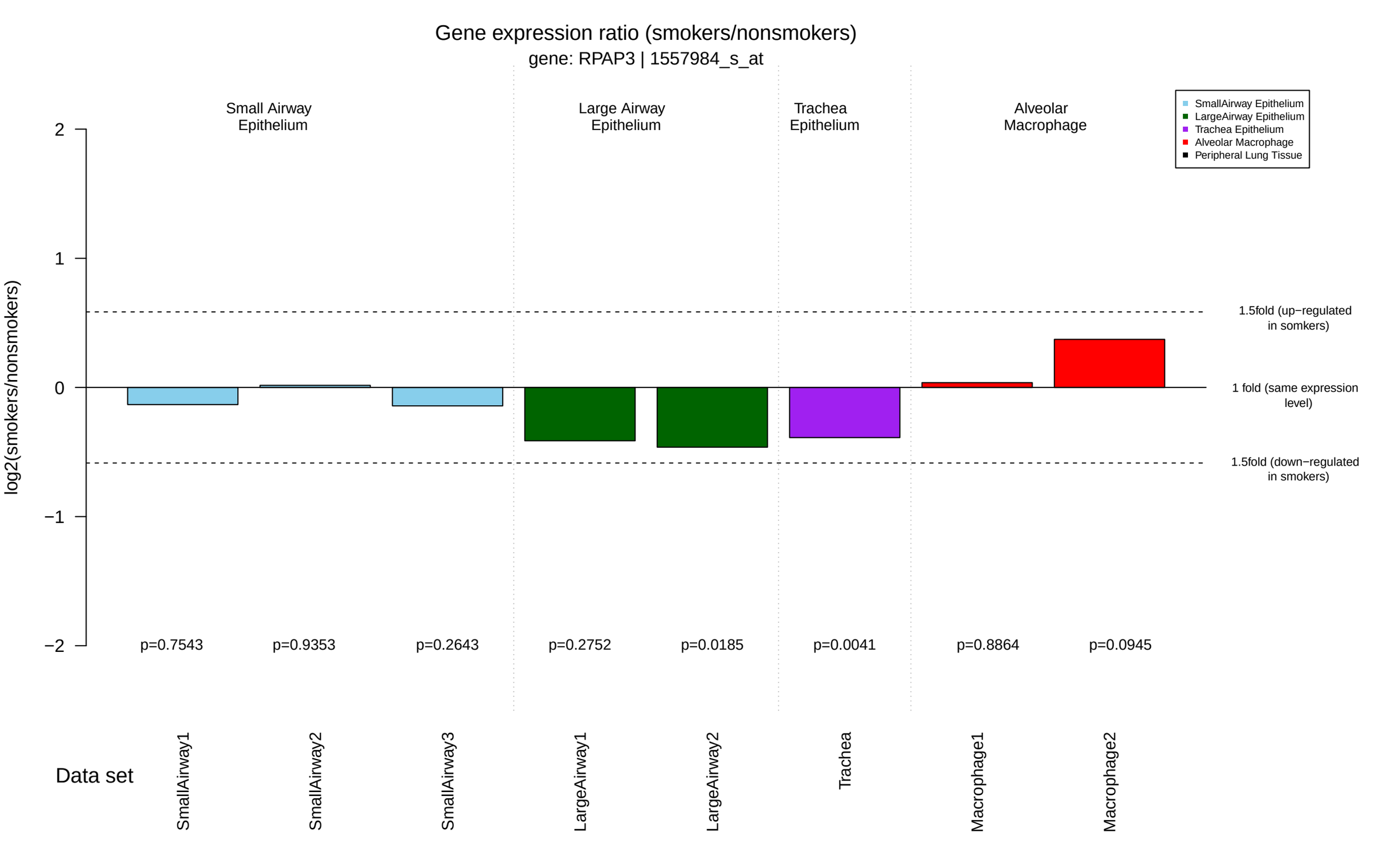
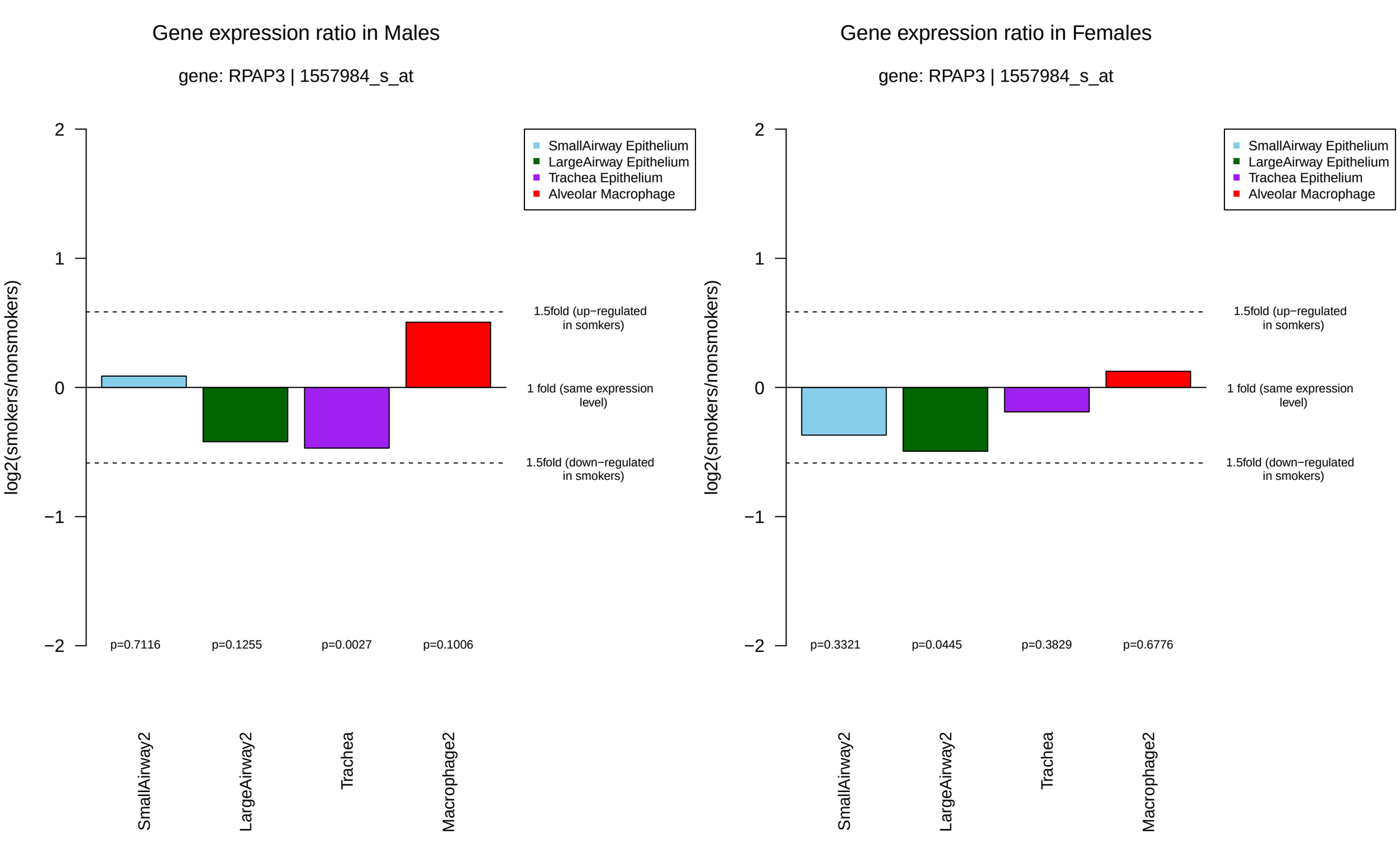
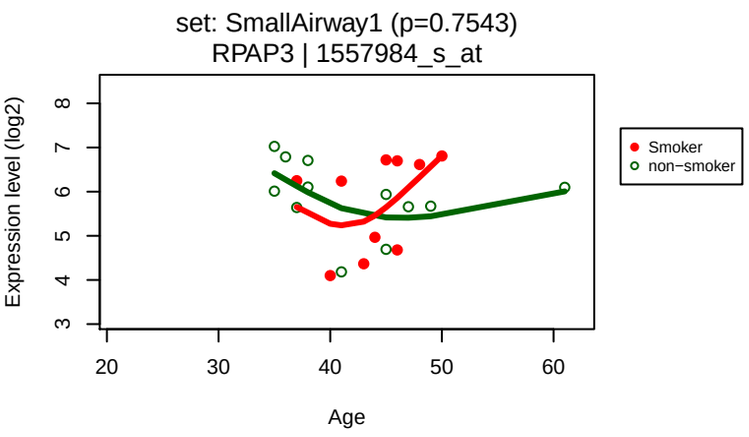
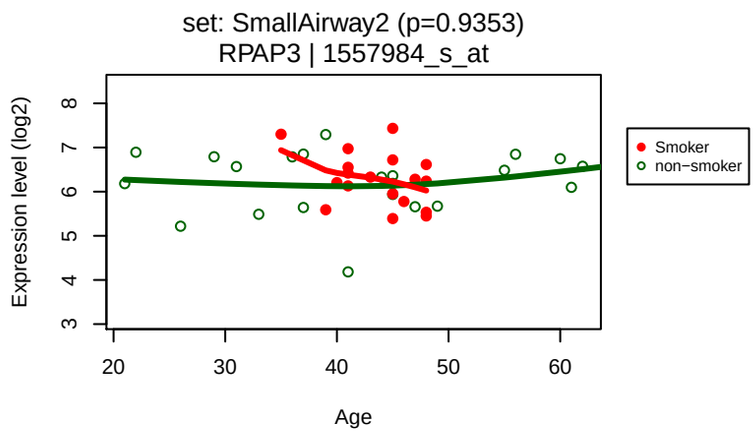
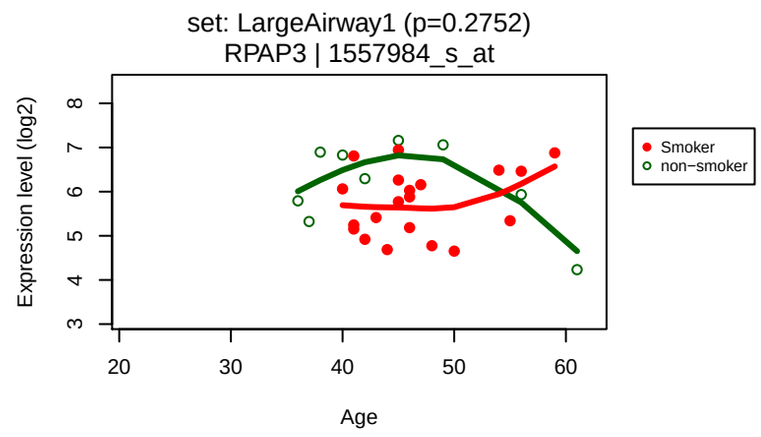

Gene expression of smokers and nonsmokers
gene: RPAP3 | 1557984_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.7543
p=0.9353
p=0.2643
p=0.2752
p=0.0185
p=0.0041
p=0.8864
p=0.0945
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

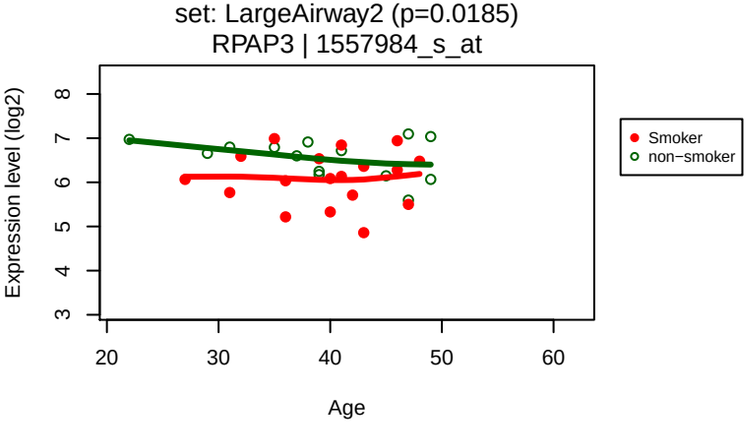

Gene expression ratio (smokers/nonsmokers)
gene: RPAP3 | 1557984_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7543
p=0.9353
p=0.2643
p=0.2752
p=0.0185
p=0.0041
p=0.8864
p=0.0945
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

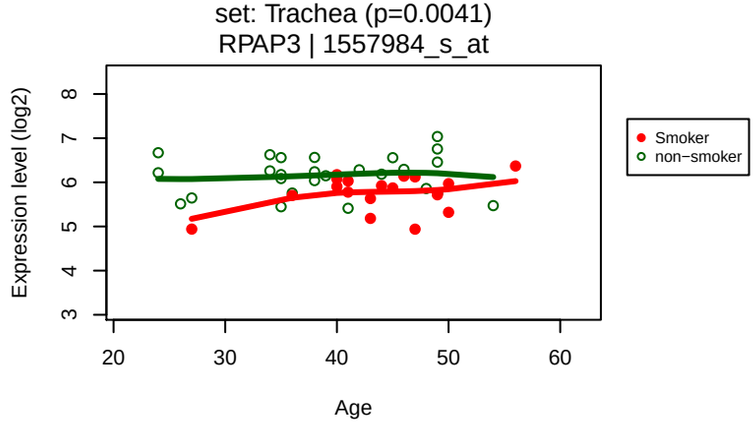

Gene expression ratio in Males
gene: RPAP3 | 1557984_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7116
p=0.1255
p=0.0027
p=0.1006
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RPAP3 | 1557984_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3321
p=0.0445
p=0.3829
p=0.6776
SmallAirway2
LargeAirway2
Trachea
Macrophage2

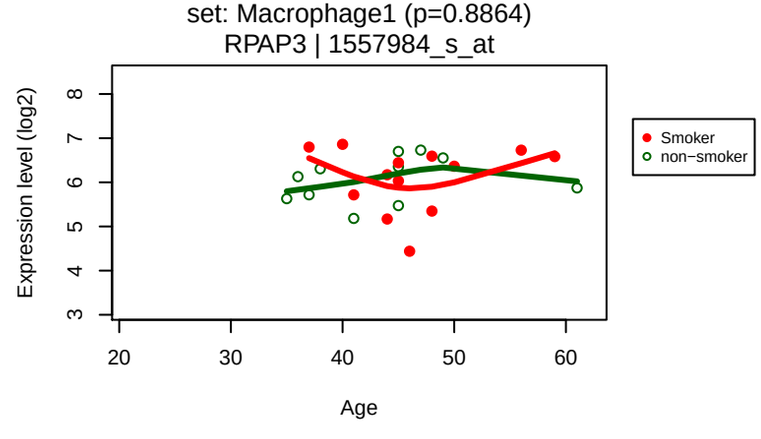

set: SmallAirway1 (p=0.7543)
RPAP3 | 1557984_s_at
Expression level (log2)
Age
Smoker
non−smoker

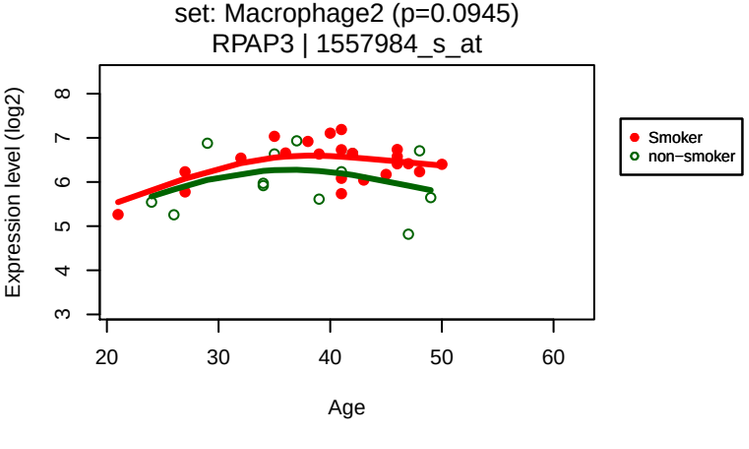

set: SmallAirway2 (p=0.9353)
RPAP3 | 1557984_s_at
Expression level (log2)
Age
Smoker
non−smoker

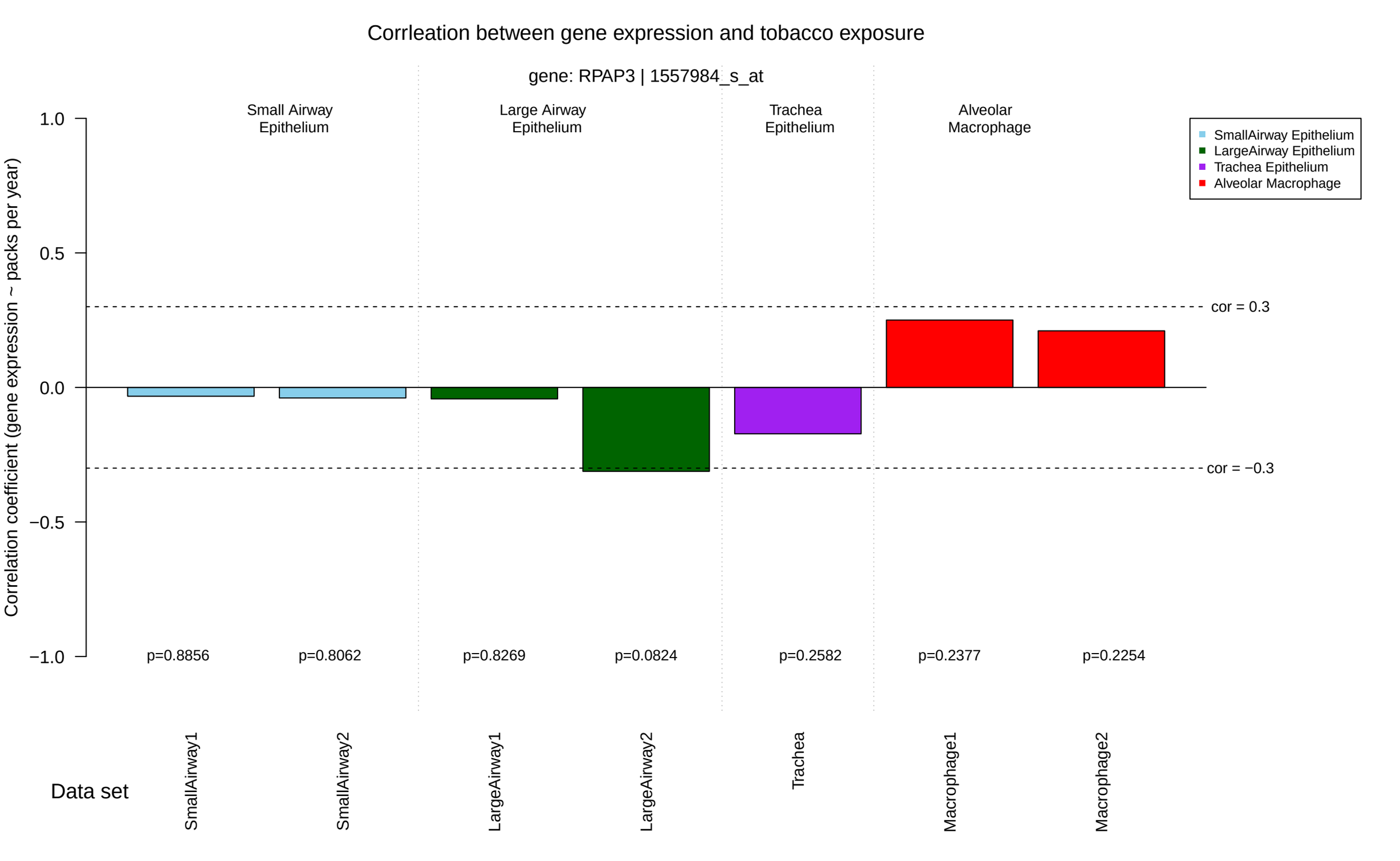

set: LargeAirway1 (p=0.2752)
RPAP3 | 1557984_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0185)
RPAP3 | 1557984_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0041)
RPAP3 | 1557984_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8864)
RPAP3 | 1557984_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0945)
RPAP3 | 1557984_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RPAP3 | 1557984_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8856
p=0.8062
p=0.8269
p=0.0824
p=0.2582
p=0.2377
p=0.2254
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2